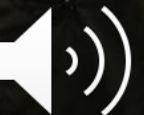


ECON1010

Introductory Microeconomics

Tutorial 5 | Elasticity



This Week in ECON1010

CML 2 is due **tomorrow** 4.00pm March 31

CML 3 opens this week 9.00am April 1

CLEAR-JE 2 is due **this Thursday** 4.00pm April 02

Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable					
Semester 1 2026					
Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- Email
- 1300 275 870 (8.30am-5pm)
- Building 42, St Lucia
- Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

UQ Campus Security
07 3365 3333 (24/7)

Off-campus emergency
000 (24/7)

Crisis support

For urgent mental health support:

UQ Counselling and Crisis Line
1300 851 998 (24/7)

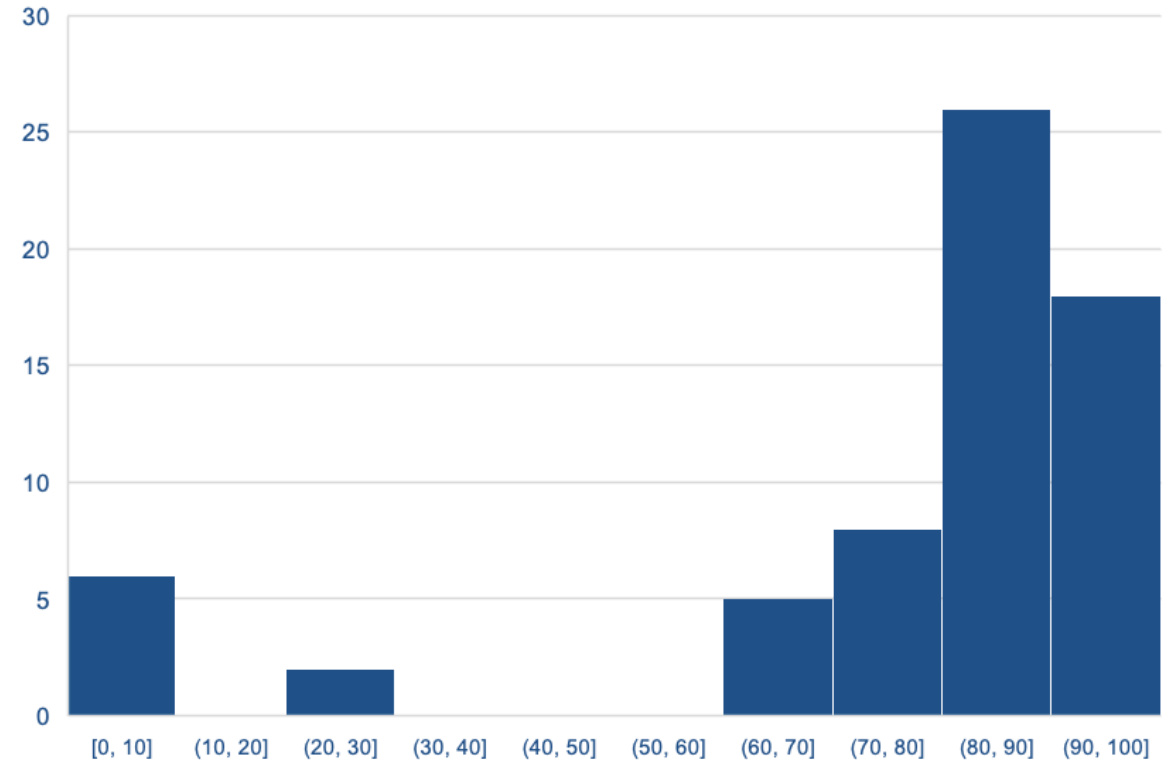
Text a Crisis Counsellor
0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

CLEAR-JE Feedback

- **Context** – was done largely well across the board – keep it simple!
- **Content** – most students achieved full marks. Remember to reference lectures in brackets and use dot points.
- **Assumptions** – students who lost marks here did so because they critiqued assumptions (do not explain why they are incorrect here).
- Use language such as “Firstly...” and “Secondly...”
- **Critique** – this section is about demonstrating deep understanding of the lecture content. Be extensive to show through engagement and comprehension.
- Mirror earlier language e.g. “Firstly, I realised my assumption blah blah blah”
- **Reflecting** – most students did well. Remember to evaluate strengths and limits of AI, and the real world application should be a clear discussion (approx. 3 sentences).

CLEAR-JE 1



Tutorial 4 (9.00am)	
Mean	85.90
Standard Error	2.67
Median	90
Mode	90

Tutorial 9 (10.00am)	
Mean	91.25
Standard Error	1.98
Median	90
Mode	90

Elasticity

Question 1ab

a) In your own words, what is:

- i) Price elasticity
- ii) Unit elasticity
- iii) Inelastic demand

b) Consider the following questions:

- i) What factors can cause the elasticity of demand to change?
- ii) What does the value of the elasticity represent?
- iii) If revenue increases with higher prices, is the good considered to be elastic or inelastic?
- iv) Why does the elasticity of demand for a good decline as we move down along a straightline demand curve? (Hint: look at formula)

Discuss with your table group, choose one person to share

Point Price Elasticity of Demand (or Supply)

formula:

$$e_d = \frac{P_i}{Q_i} * \frac{1}{\text{slope of demand curve}}$$

Mid-point Price Elasticity of Demand (or Supply)

formula:

$$e_d = \frac{\Delta Q / [(Q_A + Q_B)]}{\Delta P / [(P_A + P_B)]}$$

$$e_d = \frac{\Delta Q / Q_{AVERAGE}}{\Delta P / P_{AVERAGE}}$$

Question 1a

i) Price elasticity

Price elasticity is a measurement of how quantity (supplied or demanded) reacts to a change in price. It's a valuable analysis tool for predicting sales at different price levels.

ii) Unit elasticity

Unit elasticity is where price elasticity = 1. That means a 1% change in price will result in a 1% change in quantity (demanded or supplied – depending on which elasticity is being referred to).

iii) Inelastic demand

Inelastic demand is where price elasticity of demand is less than 1. This means quantity demanded is not very dependent on price. Obvious examples of inelastic products can be essential goods or goods that are addictive. This could be smoking (addiction) or water (essential): an increase in price doesn't greatly affect quantity demanded.

Inelastic
“I need it”

perfectly
inelastic

$$PED = 0$$

inelastic

$$PED < 1$$

unit elastic

$$PED = 1$$

elastic

$$PED > 1$$

perfectly
elastic

$$PED = \infty$$

Quantity **less**
responsive to price
change

Quantity **more**
responsive to price
change



Question 1b

i) What factors can cause the elasticity of demand to change?

1. **Availability of close substitutes** – more substitutes allow customers to change products.
2. **Time involved from the time of the price change** – more time allows new suppliers to enter the market with cheaper products or substitutes, as well as allows consumers to change what complement goods they buy.
3. **Essentials verses luxuries** – people will continue purchasing essential goods (eg water) but may choose to stop buying luxuries (eg coconut water) if the price rises.
4. **The percentage of a consumer's budget spent on the product (or service).**

Therefore, quantity demanded is more sensitive to price changes (in % terms) for goods that make up a larger proportion of your budget (and so are more elastic).

ii) What does the value of the elasticity represent?

For a 1% change in the price, there will be a (value of elasticity) % change in the quantity.

Example 1: price elasticity of demand = 3, a 3% increase in price will result in a 9% reduction in quantity demanded.

Example 2: price elasticity of supply = 0.5, a 5% increase in price will result in a 2.5% increase in quantity supplied.

Question 1b

iii) If revenue increases with higher prices, is the good considered to be elastic or inelastic?

Inelastic. As revenue = price * quantity, if price rises greatly and quantity demanded falls to a lesser extent, a producer will receive higher revenue.

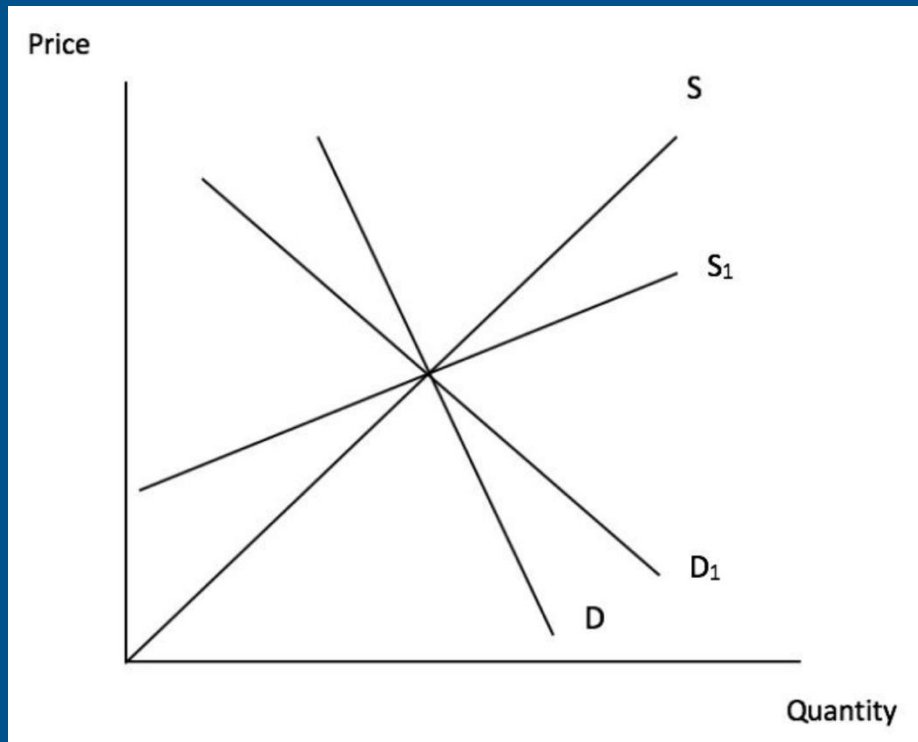
iv) Why does the elasticity of demand for a good decline as we move down along a straightline demand curve? (Hint: look at formula)

In the price point formula, it's apparent as the lower P_i is, the lower e_d . Alternatively, the higher Q_i is (as we move further down the demand curve) the lower e_d is.

$$e_d = \frac{P_i}{Q_i} * \frac{1}{\text{slope of demand curve}}$$

Question 1b

c) Draw a demand and supply graph (with labelled axes) with two demand and supply curves with different slopes. Which of the demand and supply curves are relatively more elastic?



Inelastic
“**I** need it”



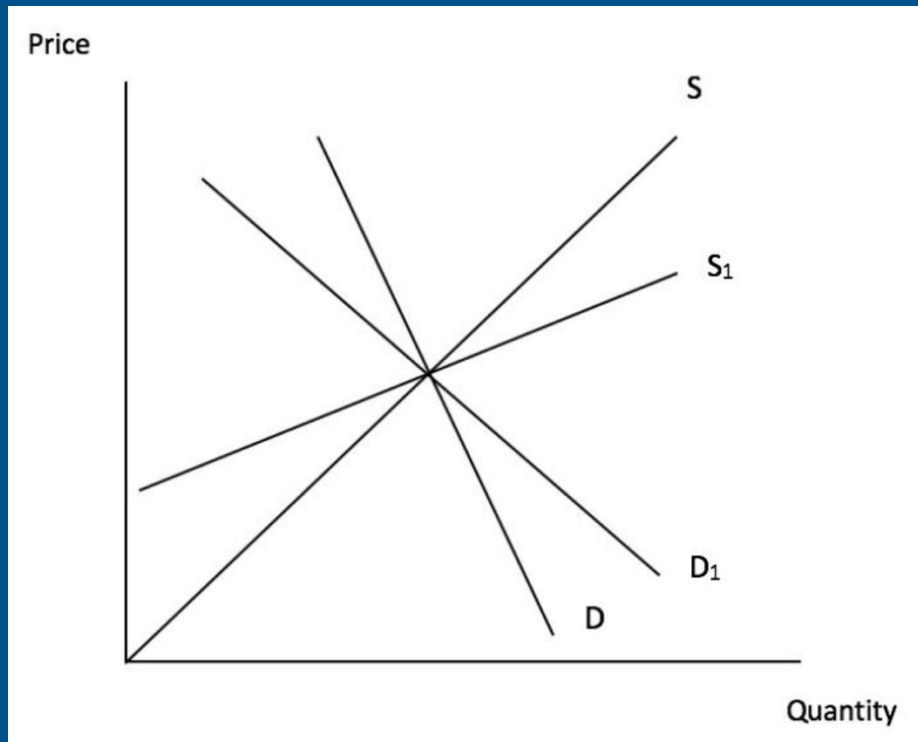
Elastic
“**E**asy without it”



S_1 and D_1 are relatively more elastic as the quantity demanded / supplied respond more to changes in price.

Question 1b

c) Draw a demand and supply graph (with labelled axes) with two demand and supply curves with different slopes. Which of the demand and supply curves are relatively more elastic?



S_1 and D_1 are relatively more elastic as the quantity demanded / supplied respond more to changes in price.

Inelastic
“**I** need it”



Elastic
“**E**asy without it”



Question 2a

At a price of \$4.50 per cappuccino, a coffee shop expects to sell 120 cappuccinos per day. At a price of \$5.50, quantity demanded of cappuccinos falls to 80.

- i) What is the price elasticity of demand for cappuccinos using the mid-point formula?
- ii) What does your answer tell you about consumers' coffee-drinking habits?
- iii) Over time, do you expect the price elasticity for cappuccinos to increase, decrease or stay the same?

$$e_d = \frac{P_i}{Q_i} * \frac{1}{\text{slope of demand curve}}$$

$$e_d = \frac{\Delta Q / Q_{AVERAGE}}{\Delta P / P_{AVERAGE}}$$



Question 2a

i) What is the price elasticity of demand for cappuccinos using the mid-point formula?

Price elasticity of demand:

$$e_d = \frac{\Delta Q / Q_{AVERAGE}}{\Delta P / P_{AVERAGE}}$$

$$e_d = \frac{\frac{120-80}{100}}{\frac{\$5.50-\$4.50}{\$5}} = \frac{4/10}{1/5} = 2$$

It is common to see Price Elasticity of Demand (PED) as a positive number (absolute number) with the negative sign disregarded, as PED is almost always negative. In more advanced economics, elasticity can compare different variables than just price and quantity. In these cases, the positive/negative sign is important.

ii) What does your answer tell you about consumers' coffee-drinking habits?

A value of more than 1 indicates that demand is elastic. The proportional decrease in quantity demanded is more than the proportional rise in price. Consumers are relatively sensitive to the price rise, which would be expected of products with many close substitutes (flat white, latte etc). As the price of cappuccinos has risen, coffee-drinkers are likely to switch to different coffee types.

Question 2a

iii) Over time, do you expect the price elasticity for cappuccinos to increase, decrease or stay the same?

In general, we expect the price elasticity to increase over time (become more elastic). Time allows consumers the opportunity to adjust their consumption habits and switch from cappuccinos to an alternative product.



Question 2b

The schedule below the number of croissants customers purchase from all bakeries across a city each day at various prices.

- i) Graph the daily demand curve for croissants in the city.
- ii) Calculate the price elasticity of demand for croissants, when the price is \$3 per croissant, using the point elasticity of demand formula. Is demand elastic or inelastic?
- iii) If the price rises from \$3 per croissant to \$4 per croissant, what does economic theory tell you will happen to revenue based on the demand elasticity?
- iv) Calculate the revenues at \$3 per croissant and \$4 per croissant to verify if your answer in iii) is correct.
- v) If you were incorrect in iii), can you think of why?

Price (\$/croissant)	Number of croissants purchased per day
6	0
5	3,000
4	6,000
3	9,000
2	12,000
1	15,000
0	18,000

Remember – as we move down the demand curve, elasticity decreases



Question 2b

- i) Graph the daily demand curve for croissants in the city.
- ii) Calculate the price elasticity of demand for croissants, when the price is \$3 per croissant, using the point elasticity of demand formula. Is demand elastic or inelastic?

Price elasticity of demand is calculated as $(1/\text{slope}) * (P/Q)$.

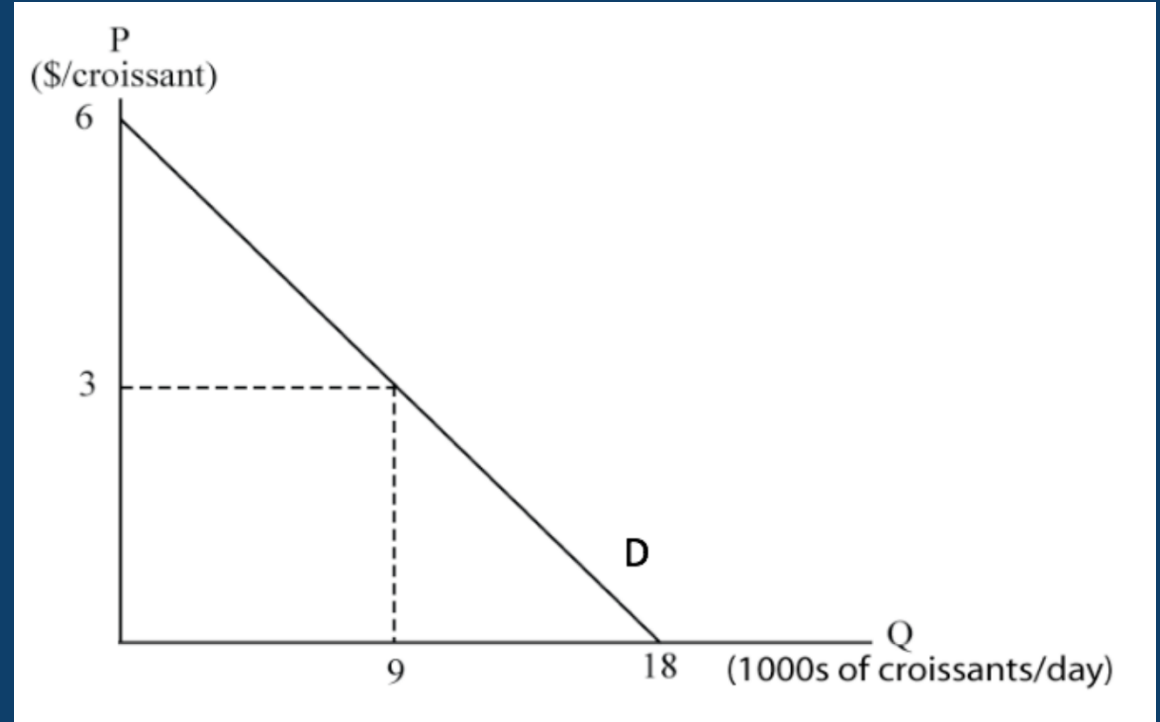
When $P = \$3$ each, $Q = 9$, the slope is thus $(6/18) = 1/3$ so:

$$\text{PED} = 1/(1/3) * P/Q$$

$$\text{PED} = 3 * (P/Q)$$

$$\text{PED} = 3 * (3/9) = 3 * (1/3) = 1$$

So, demand is unit elastic ($\text{PED} = 1$), meaning quantity demanded changes proportionally with price.



$$e_d = \frac{P_i}{Q_i} * \frac{1}{\text{slope of demand curve}}$$

Question 2b

iii) If the price rises from \$3 per croissant to \$4 per croissant, what does economic theory tell you will happen to revenue based on the demand elasticity?

Decrease. As the price rises from \$3 to \$4, the Mid-Point Price Elasticity of Demand is more accurate than PED at \$3. As PED increase further up the demand curve [see Q1 b) iv)], the demand will be more elastic at \$3.50 than \$3. Because demand was unit elastic at \$3, it will be elastic at \$3.50. Since demand is elastic, already small quantities reduce to even smaller quantities at a higher price. This reduction in quantity is more significant than any increase in price in the elastic region of the demand curve. As a result, when applying the formula $\text{Revenue} = \text{quantity} \times \text{price}$, revenues are found to decrease.

iv) Calculate the revenues at \$3 per croissant and \$4 per croissant to verify if your answer in iii) is correct.

At \$3, quantity demanded is 9,000 so $TR = P \times Q = 3 \times 9,000 = \$27,000$

At \$4, quantity demanded is 6,000 so $TR = P \times Q = 4 \times 6,000 = \$24,000$

Revenue will fall from \$27 000 to \$24 000.

v) If you were incorrect in iii), can you think of why?

Sometimes students believe the unit-elasticity at \$3 is the correct elasticity between \$3 and \$4 and therefore estimate that revenues will remain constant – this is not true. Elasticity increases at higher prices (higher up the demand curve), resulting in price elasticity of demand greater than 1 between \$3 and \$4.

Question 2c

Consider two markets:

- Market 1: Dell products (consists of electronic hardware).
- Market 2: Electronic hardware (all companies).

Which market's demand is more elastic? Why?

Market 1 is more elastic. Dell's products have many close substitutes for their electronic hardware, thereby making Dell's products more elastic. Market 2, the class of electronic hardware (including Dell products) has no close substitutes at all. Therefore consumers will pay very high prices for the goods in this market.



Question 3

a) Reflect on your own personal situation:

Think of an item/event/activity you have bought this week (pizza, fuel, music festival tickets)

- i) Is it an essential or luxury product?
- ii) Is the elasticity of demand for this product generally elastic or inelastic?
- iii) Which of the factors affecting elasticity of demand will greatly affect this product?



Thank You

Next week: Mid-Semester Break!

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